Computer Organization (CSD206): Tutorial 1

For this tutorial, you will solve questions based on Number System.

1. Convert the following decimal numbers to their binary equivalents:
   a) 64
   b) 111
   c) 34.75
   d) 25.25

2. Convert the following binary numbers to their decimal equivalents:
   a) 001100
   b) 000011
   c) 11100.011

3. Convert the following hexadecimal numbers to their binary equivalents:
   a) 1C
   b) 239.4

4. Convert the following binary number to its hexadecimal equivalent: 110101.011001

5. Convert the following hexadecimal number to its decimal equivalent: EBA.C

6. Convert the following decimal number to its hexadecimal equivalent: 2560

7. Order the numbers $(1.1)_2$, $(1.4)_{10}$, and $(1.5)_{16}$ from smallest to largest.

8. Perform the indicated base conversions:
   a) 54_8 to base 5
   b) 312_4 to base 7